

## ☑ SAFETY DATA SHEET

**GreenX™**

Hydrogen Fuel Assist Retrofit

- ▶ DG Sets: 20 kVA to 4000 kVA | Any Make | Any Age
- ▶ India Patent No: 582824 | Filed: 26/02/2024
- ▶ Application No: 202421013845

Doc No: SGT-GX-SDS-001 | Rev: 1.0 | March 2026

**SGT HydroEdge Pvt. Ltd.**

Industrial Decarbonisation Technologies

- ✓ ISO 9001:2015 (TUV Austria)
- ✓ RoHS | CE | UL | FCC | IEC | EMC
- ✓ NABL Accredited Lab Tested
- ✓ TRL 8–9 Proven Technology

contact@sgthydroedge.com  
www.sgthydroedge.com

**3–15%**

Diesel Reduction

**~80%**

PM Reduction

**~70%**

NOx Reduction

**~87%**SO<sub>2</sub> Reduction**3.5–18%**tCO<sub>2</sub>e Reduction**8**

Safety Layers

**1 SECTION 1: PRODUCT IDENTIFICATION**

|                               |                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Product Name</b>           | GreenX™ Hydrogen Fuel Assist Retrofit                                                            |
| <b>Product Code</b>           | SGT-GreenX Series                                                                                |
| <b>Product Type</b>           | On-Demand HHO Gas Generation & Injection System for Diesel Generator Engines                     |
| <b>Application Range</b>      | Compatible with DG sets: 20 kVA — 4000 kVA   Any make   Any age                                  |
| <b>Technology Readiness</b>   | TRL 8–9   Commercially Deployed                                                                  |
| <b>Patent Status</b>          | India Patent No: 582824   Application No: 202421013845   Filed: 26/02/2024                       |
| <b>Manufacturer</b>           | SGT HydroEdge Private Limited                                                                    |
| <b>Factory Address</b>        | Gat No. 357/86, Plot No. 14, Waghjainagar, Kharabwadi, Chakan, Pune – 410501, Maharashtra, India |
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| <b>Document Number</b>        | SGT-GX-SDS-001   Rev 1.0   March 2026                                                            |

## 2 SECTION 2: PRODUCT DESCRIPTION & WORKING PRINCIPLE

### 2.1 What is GreenX™?

GreenX™ is a patent-applied, on-demand HHO gas generation system designed as a bolt-on retrofit for diesel generators. It does NOT store any HHO gas. Instead, it electrolyzes water in real time and immediately injects the generated HHO gas into the engine's air intake manifold, improving combustion efficiency and significantly reducing harmful exhaust emissions.

#### IMPORTANT: GreenX™ Generates HHO Gas — Not Separate H<sub>2</sub> or O<sub>2</sub>

GreenX™ generates HHO gas — a combined mixture of Hydrogen and Oxygen produced simultaneously by water electrolysis. The system does NOT separate or independently store Hydrogen (H<sub>2</sub>) or Oxygen (O<sub>2</sub>). This combined HHO gas is injected directly into the engine air intake in its mixed form and is fully consumed during combustion. This combined generation approach is an additional inherent safety advantage.

### 2.2 How It Works

- Water (H<sub>2</sub>O) is fed into the electrolytic wet cell from the auxiliary reservoir.
- The wet cell uses electrical energy from the DG battery to split water into HHO gas via electrolysis.
- The HHO gas is electronically metered and immediately fed into the engine's air intake manifold.
- Hydrogen's high flame speed accelerates diesel combustion, reducing ignition delay and improving thermal efficiency.
- Result: More complete diesel burn, lower fuel consumption, and dramatically reduced exhaust emissions.

### 2.3 Validated Performance (NABL Accredited Lab)

|                              |                                                                  |
|------------------------------|------------------------------------------------------------------|
| Diesel Fuel Reduction        | 3% to 15% (average 8.5%) — dependent on load                     |
| Particulate Matter (PM)      | 75% to 80% reduction (up to 95% at certain loads)                |
| NO <sub>2</sub> Reduction    | 50% to 72%                                                       |
| SO <sub>2</sub> Reduction    | 62% to 87%                                                       |
| CO & HC Reduction            | 60% to 80%                                                       |
| tCO <sub>2</sub> e Reduction | Estimated 3.5% to 18% (incl. N <sub>2</sub> O avoidance benefit) |
| Tested on                    | 380 kVA and 1500 kVA   Results consistent 125 kVA–1500 kVA       |

## 3 SECTION 3: REGULATORY COMPLIANCE & CERTIFICATIONS

### 3.1 AIS-157A Applicability Statement

#### AIS-157A Does NOT Apply to GreenX™

AIS-157A is applicable to OEM hydrogen fuel cell vehicles with compressed hydrogen storage systems operating above 30 bar (Class 0 components). GreenX™ is a retrofit, on-demand HHO generation unit with NO gas storage, operating well below 30 bar. As per AIS-157A: hydrogen fuel system components with nominal working pressure up to 30 bar are explicitly exempted from type approval requirements. GreenX™ falls entirely outside the scope of AIS-157A. No type approval under this standard is required.

- AIS-157A applies exclusively to OEM vehicles with compressed hydrogen storage tanks.

- GreenX™ generates and immediately consumes HHO gas — zero storage at any stage.
- Operating pressure of GreenX™ is below 30 bar — classified as non-Class 0.
- Hydrogen fuel system components with nominal working pressure ≤30 bar are explicitly exempted.
- GreenX™ connects only to the air intake — no engine modification or high-pressure lines involved.

### 3.2 Certifications

| Certification        | Certificate No. | Issued By                              | Valid Until |
|----------------------|-----------------|----------------------------------------|-------------|
| ISO 9001:2015        | 20100243020317  | TUV Austria GMBH                       | 2027-09-09  |
| RoHS                 | RM-50365/0125   | PQC — Paramount Quality Certifications | 16.01.2028  |
| EMC                  | EM-50366/0125   | PQC — Paramount Quality Certifications | 16.01.2028  |
| CE (GPSD 2001/95/EC) | CM-50362/0125   | PQC — Paramount Quality Certifications | 16.01.2028  |
| UL Guidelines        | UM-50364/0125   | PQC — Paramount Quality Certifications | 16.01.2028  |
| FCC Part 15A & 15B   | FM-50363/0125   | PQC — Paramount Quality Certifications | 16.01.2028  |
| IEC Compliance       | CM-50367/0125   | PQC — Paramount Quality Certifications | 16.01.2028  |

## 4

### SECTION 4: BUILT-IN SAFETY FEATURES — DETAILED DESCRIPTION



All 8 safety features are fully automatic. GreenX™ activates, shuts down, and restarts without any manual intervention, ensuring safe and reliable 24/7 operation.



SF-01

#### No HHO Gas Storage — Inherently Safe by Design

**What it does:** GreenX™ does NOT store any HHO gas at any point in the system.

**How it works:** HHO gas is generated on-demand via electrolysis and immediately injected into the engine air intake. Generation and consumption are simultaneous.

**Safety benefit:** Zero gas accumulation. No pressurized flammable gas buildup. No explosion hazard. No high-pressure vessel required.

**Significance:** This is the most fundamental safety feature — the system is inherently safe at the architecture level.



SF-02

#### Engine-Synchronized Auto Start / Auto Shut-Off

**What it does:** GreenX™ automatically starts when the diesel engine starts and shuts down the moment the engine stops.

**How it works:** The system monitors the engine start signal directly. Power to the electrolytic cell is enabled only when the engine is confirmed running.

**Safety benefit:** Eliminates any possibility of HHO generation when the engine is off. No uncontrolled gas generation during idle or shutdown.

**Operator action:** None required. Fully automatic operation.

|                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>SF-03   | <b>Battery Voltage Threshold Protection (DG Battery Safety)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                                              | <p><b>What it does:</b> Protects the DG set battery from drain, ensuring the engine can always start reliably.</p> <p><b>24V DG System (Dual Battery):</b> GreenX™ draws power only when battery voltage is above 23.8V. Auto-stops if voltage drops to or below 23.8V.</p> <p><b>12V DG System (Single Battery):</b> GreenX™ draws power only when battery voltage is above 11.5V. Auto-stops if voltage drops to or below 11.5V.</p> <p><b>Auto recovery:</b> System restarts automatically when voltage recovers above the respective threshold. No manual intervention needed.</p> <p><b>Safety benefit:</b> Prevents battery drain, ensures DG starting reliability in all conditions, protects battery health and lifespan.</p> |
| <br>SF-04   | <b>Gas Blockage / Back-Pressure Shut-Off Protection</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                              | <p><b>What it does:</b> Protects against unsafe pressure buildup if the HHO delivery pipe becomes blocked or kinked.</p> <p><b>How it works:</b> The system continuously monitors line pressure in the HHO delivery pathway. If pressure reaches 0.1 bar above normal operating pressure (indicating a blockage), the system automatically shuts down.</p> <p><b>Auto recovery:</b> Once the blockage is cleared and pressure returns to normal, the system automatically restarts. No manual action needed.</p> <p><b>Safety benefit:</b> Prevents potential pipe rupture, HHO accumulation, or unsafe pressure conditions in the delivery line.</p>                                                                                 |
| <br>SF-05 | <b>Water Level Low — Automatic Shut-Off &amp; Auto-Restart</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                                              | <p><b>What it does:</b> Prevents the electrolytic cell from running dry, avoiding overheating or damage to electrolytic plates.</p> <p><b>How it works:</b> A water level sensor in the auxiliary reservoir monitors water level continuously. If water drops below the minimum safe level, the system automatically shuts off.</p> <p><b>Auto recovery:</b> Once the operator fills the reservoir to adequate level, the system automatically restarts. No manual reset required.</p> <p><b>Operator action:</b> Refill the water reservoir. Restart is fully automatic after refill.</p>                                                                                                                                            |
| <br>SF-06 | <b>DM Water Requirement &amp; Integrated Water Filtration System</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                                              | <p><b>Primary requirement:</b> GreenX™ requires Demineralized (DM) water for optimal electrolysis and to prevent mineral scaling on electrolytic plates.</p> <p><b>Built-in safety backup:</b> If an operator fills the reservoir with normal (non-DM) water, the integrated water filtration system automatically converts it to DM-quality water before use.</p> <p><b>Filter capacity:</b> The internal filter supports up to 300 liters of normal water before requiring replacement.</p>                                                                                                                                                                                                                                         |

**Recommendation:** SGT strongly recommends using only DM water to maximize filter life, prevent scale buildup, and ensure peak system performance.



### Electronics Overheating Protection — Thermal Shut-Off

**What it does:** Protects control electronics from damage due to excessive operating temperature.

**How it works:** A thermal sensor continuously monitors the temperature of the control electronics/PCB. If temperature reaches or exceeds 80°C, the system automatically shuts down.

**Auto recovery:** System restarts automatically once electronics cool below the safe threshold. No manual reset needed.

**Safety benefit:** Prevents thermal damage to electronic components, ensuring long operational life even in high ambient temperature environments.



### Smart IoT Monitoring, Remote Alarm & Malfunction Auto Shut-Off

**IoT platform:** GreenX™ is integrated with SGT GreenVision™ IoT monitoring platform for continuous smart oversight.

**What it monitors:** Real-time system status, fuel consumption, DG load profile, emission signatures, water level, voltage, temperature, and pressure.

**Malfunction response:** If any abnormal parameter is detected, the system automatically shuts down immediately.

**Alarm system:** Alarm notifications are triggered on the SGT GreenVision™ dashboard and sent to the operator remotely. All events are logged with timestamp and event type.

**Manual monitoring:** Operators can view live data, historical logs, and alarm history from the GreenVision™ dashboard at any time from anywhere.

## 5 SECTION 5: SAFETY FEATURES QUICK REFERENCE TABLE

| Code  | Safety Feature                 | Trigger Condition                    | Auto Response                                   |
|-------|--------------------------------|--------------------------------------|-------------------------------------------------|
| SF-01 | No HHO Gas Storage             | Inherent — design level              | Zero gas accumulation at all times              |
| SF-02 | Engine-Synced Auto Start/Stop  | Engine ON/OFF signal                 | Auto-start with engine; auto-stop on engine off |
| SF-03 | Battery Voltage Threshold      | 24V: ≤23.8V / 12V: ≤11.5V            | Auto-stop; auto-restart when voltage recovers   |
| SF-04 | Gas Blockage Pressure Shut-Off | Line pressure > 0.1 bar above normal | Auto-stop; auto-restart on clearance            |
| SF-05 | Water Level Low Shut-Off       | Water sensor detects low level       | Auto-stop; auto-restart when water refilled     |
| SF-06 | DM Water Filtration Backup     | Non-DM water in reservoir            | Integrated filter converts up to 300L           |
| SF-07 | Electronics Overheating        | PCB temp ≥80°C                       | Auto-stop; auto-restart on cool-down            |
| SF-08 | IoT Malfunction Monitoring     | Any abnormal parameter detected      | Auto-stop + remote alarm on GreenVision™        |

## 6

**SECTION 6: HANDLING, INSTALLATION & OPERATING PRECAUTIONS****6.1 Pre-Installation Checks**

- Ensure the DG set is switched OFF and the battery is disconnected before beginning installation.
- Verify that the GreenX™ unit model matches the DG set's kVA rating.
- Inspect all supplied components (wet cell, HHO tubing, control unit, water reservoir) for transit damage.
- Identify the air intake manifold connection point on the DG engine before starting.

**6.2 Installation Guidelines**

- Installation is completed in under 4 hours by a trained SGT technician or authorized installer.
- The system connects only to the engine's air intake manifold — no engine disassembly or modification required.
- Electrical connections draw from the existing 24V or 12V DG battery system — no separate power supply needed.
- Route HHO delivery tubing away from hot engine surfaces and moving parts; use all provided clamps and brackets.
- Fill the water reservoir with Demineralized (DM) water before first operation.

**6.3 Water Handling**

- Use Demineralized (DM) water as the primary consumable. Integrated filter provides backup for non-DM water (up to 300L).
- Check the water level before each DG operating session.
- System auto-shuts on low water and auto-restarts after refill — no manual reset needed.

**6.4 Operational Precautions**

- Do not tamper with, block, or kink the HHO gas delivery tubing during operation.
- Do not attempt to collect, store, or accumulate HHO gas from the system output.
- Maintain adequate ventilation in the DG room or enclosure at all times.
- In case of unusual smell, smoke, or visible abnormality, stop the DG set immediately and contact SGT: [contact@sgthydroedge.com](mailto:contact@sgthydroedge.com)

## 7

**SECTION 7: STORAGE, MAINTENANCE & DISPOSAL****7.1 Storage Guidelines**

- Store the GreenX™ unit in a cool, dry, and well-ventilated location suitable for outdoor industrial equipment.
- Drain the water reservoir if storing for more than 7 consecutive days to prevent algae or mineral buildup.
- Keep stored unit away from flammable materials, strong oxidizing agents, and corrosive environments.
- Protect electrical connections from moisture and condensation during extended storage.

**7.2 Routine Maintenance Schedule**

|                            |                                                                        |
|----------------------------|------------------------------------------------------------------------|
| <b>Before each session</b> | Check and top up DM water reservoir                                    |
| <b>Monthly</b>             | Inspect HHO delivery tubing for wear, kinks, cracks, or loose fittings |

|                                |                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------|
| <b>Quarterly</b>               | Check all electrical connections for corrosion; inspect wet cell electrolytic plates |
| <b>After 300L non-DM water</b> | Replace water filtration cartridge                                                   |
| <b>Annually</b>                | Full service by SGT-certified technician under Annual Maintenance Contract (AMC)     |

### 7.3 Disposal & Environmental Compliance

- GreenX™ electronics comply with RoHS Directive — free of Lead, Mercury, Cadmium, Chromium VI, PBB, and PBDE.
- Dispose of electronic components per applicable local e-waste (WEEE) regulations.
- Neutralize and dispose of wet cell electrolyte solution per local chemical waste regulations.
- Contact SGT HydroEdge at [contact@sgthydroedge.com](mailto:contact@sgthydroedge.com) for authorized disposal or recycling guidance.

## 8 SECTION 8: ENVIRONMENTAL & REGULATORY INFORMATION

|                                                 |                                                                                                                                                           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RoHS Compliance</b>                          | Fully compliant (Cert. No: RM-50365/0125) — free of all RoHS-restricted hazardous substances                                                              |
| <b>N<sub>2</sub>O Emissions</b>                 | Zero — GreenX™ uses NO Oxidation Catalysts. Unlike SCR/DOC retrofits, no N <sub>2</sub> O is generated.                                                   |
| <b>tCO<sub>2</sub>e Net Impact</b>              | Positive: Combined fuel savings + N <sub>2</sub> O avoidance = 3.5% to 18% effective tCO <sub>2</sub> e reduction                                         |
| <b>ESG Reporting</b>                            | Supports Scope 1 & Scope 3 GHG emission reduction reporting. All reductions are quantifiable and reportable.                                              |
| <b>CPCB / EPA Norms</b>                         | Significant PM, NO <sub>x</sub> , and CO reductions align with CPCB emission norms for DG sets.                                                           |
| <b>No Fuel Switching</b>                        | Operates with standard diesel — no blended or alternative fuel needed.                                                                                    |
| <b>N<sub>2</sub>O Advantage vs. Competitors</b> | Oxidation Catalyst retrofits generate N <sub>2</sub> O (298x CO <sub>2</sub> e factor) AND increase diesel use due to back-pressure. GreenX™ avoids both. |

## 9 SECTION 9: MANUFACTURER CONTACT & SUPPORT

|                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SGT HydroEdge Private Limited</b><br><i>Industrial Decarbonisation Technologies</i><br><ul style="list-style-type: none"> <li>Plot #14, Gat #357/86, Waghjainagar<br/>Kharabwadi, Chakan, Pune – 410501<br/>Maharashtra, India</li> </ul> | <b>Get in Touch</b><br><ul style="list-style-type: none"> <li>Email: <a href="mailto:contact@sgthydroedge.com">contact@sgthydroedge.com</a></li> <li>24/7 Emergency: +91 9049270404</li> <li>Website: <a href="http://www.sgthydroedge.com">www.sgthydroedge.com</a></li> </ul> |
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### Document Disclaimer



This Safety Data Sheet is prepared in good faith based on product design specifications, NABL-accredited test data, and regulatory knowledge as of March 2026. SGT HydroEdge Pvt. Ltd. reserves the right to update this document as product specifications or applicable regulations

evolve. Always refer to the latest version at [www.sgthydroedge.com](http://www.sgthydroedge.com).

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